FOREIGN TRADE UNIVERSITY
VIETNAM-JAPAN SCHOOL OF BUSINESS AND INNOVATION

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FINAL REPORT
ADVANCED BUSINESS ANALYTICS

HOTEL BOOKING CANCELLATION DECISION SUPPORT CITY HOTEL LISBON

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1. Strategic Problem Statement

1.1. Business Context

Scenario: City Hotel Lisbon sells perishable room inventory ahead of arrival, largely through travel agents and operators. Until the guest arrives, revenue is only expected, not secured.

Strategic issue: Revenue-management research identifies booking cancellations as a core source of demand uncertainty: they limit the accuracy of forecasts, and the usual responses, rigid cancellation policies and overbooking, can themselves damage revenue and reputation (Antonio et al., 2017). For City Hotel Lisbon, this means expected revenue and room plans cannot be fully relied on, and blanket policies are an imperfect fix. Cancellation is therefore a strategic revenue-risk problem, not just an operational event. Its scale is quantified in Section 1.2.

Cross-domain impact: One cancellation touches several domains at once, yet each is managed separately today, so no single view supports a joint decision.

Domain	Current state	Gap the project addresses
Distribution / Marketing	Cancellation risk differs sharply by channel and segment, but is not used in booking management	Segment and agent risk guides follow-up priority and group-block contracts
Customer Experience	At-risk guests are not systematically contacted	Proactive, targeted contact for high-risk bookings only
Finance	Commission, handling cost and revenue at risk are not linked	Net value of actions accounts for cost
Operations	Freed rooms are not systematically resold	Resale and capacity decisions are guided by cancellation risk

Transformation opportunity. Integrating booking, customer, cost and inventory data into one decision-support system can shift the hotel from reactive handling to risk-based, proactive booking management, optimizing revenue and room capacity.

1.2. Business Pain Points

Baseline: City Hotel Lisbon, 79,330 bookings, July 2015 - August 2017 (Antonio, Almeida & Nunes, 2019).

Pain point	Supporting data	Why it matters
Limited visibility into cancellation risk	41.7% of bookings (33,102) were cancelled or no-show. Most cancellations come weeks ahead (median 58 days before arrival); 12.0% come within 7 days (14.5% if no-shows count as 0-day notice)	Most cancellations can still be acted on weeks before arrival, but only if risk is known early. Late cancellations leave the least time to resell the room, so they are handled through a capacity buffer rather than guest contact

Inefficient allocation of follow-up resources	Travel agents and operators handle 87% of bookings. Cancellation rates differ by segment: Direct 17.3%, Online TA 37.4%, Groups 68.9%. Group and Non Refund blocks need contract-level action; individual guests need booking-level follow-up. At-risk guests are not systematically contacted (assumed current practice)	Risk varies widely across segments, so uniform follow-up wastes effort on safe bookings and misses risky ones. Hospitality churn research supports targeting only high-risk guests rather than contacting everyone (van Leeuwen et al., 2025)
Limited ability to optimize revenue and room capacity	€10.9M of €25.3M gross booked room revenue (about 43%) is tied to cancelled bookings. Resale of freed rooms is not guided by risk, and commission and handling cost are not linked to risk (assumed current practice)	The hotel cannot tell which bookings to act on, or how to adjust follow-up, pricing, capacity allocation and resale so that revenue and occupancy are optimized net of intervention cost. Revenue-management studies show that cancellation-aware allocation and pricing decisions can raise profit in simulation (Dong & Ling, 2015; Cha, 2026; Pimentel et al., 2019)

Note: €10.9M is gross booking value exposed to cancellation, not realized loss. Some of it is resold or recovered through deposits.

1.3. Stakeholders

Group	Role	Interest
General Manager, Revenue Manager	Decision-makers	Set priorities, resources and booking-management strategy; optimize revenue and capacity
Reservation Team	Operational users	Use risk information to follow up with guests and flag freed rooms to the Revenue Manager
Customer Service / CX	Operational users	Design and send guest messages; own opt-outs and CX KPIs
Finance	Decision support	Own cost and contribution assumptions; judge net value and ROI of interventions

Distribution / Sales	Operational users	Manage agents, tour operators and group-block contracts (release dates, deposits)
Data / Analytics Team	System owners	Build the model, check its calibration and monitor drift
Guests	Affected party	Experience the interventions; not direct system users

1.4. Decisions to Improve

Decision	Business question	Supported by
Risk identification	Which bookings are more likely to be cancelled?	Machine Learning
Resource prioritization	Which bookings should staff handle first when resources are limited?	Risk score + value proxy
Intervention selection	Should the hotel confirm, send a reminder or contact the guest directly?	Business rules
Revenue and capacity optimization	Which follow-up, capacity-allocation and resale actions improve revenue and occupancy at a justified cost, and would price adjustment add value?	Rule-based analysis + ROI analysis

Only the first decision is addressed directly by ML. The others need business rules, operational data and impact evaluation.

1.5. Strategic Objectives

Each objective follows from a pain point and a decision, and is backed by evidence.

	Objective	Addresses pain	Improves decision	Evidence basis
1	Identify cancellation risk at booking time, while there is still time to act	Limited visibility	Risk identification	41.7% cancellation rate; median notice 58 days; 12.0% within 7 days
2	Allocate follow-up effort to the highest-risk, highest-value bookings	Inefficient allocation	Resource prioritization; intervention selection	Segment gap from 17.3% (Direct) to 37.4% (Online TA) among individual bookings, 68.9% for Groups; threshold-based targeting (van Leeuwen et al., 2025)

	Objective	Addresses pain	Improves decision	Evidence basis
3	Optimize revenue and room capacity by supporting follow-up, capacity-allocation and resale decisions net of intervention cost, and test price adjustment as a scenario	Limited ability to optimize revenue and capacity	Revenue and capacity optimization	€10.9M exposure; profit gains in simulated revenue-management studies (Dong & Ling, 2015; Cha, 2026; Pimentel et al., 2019)

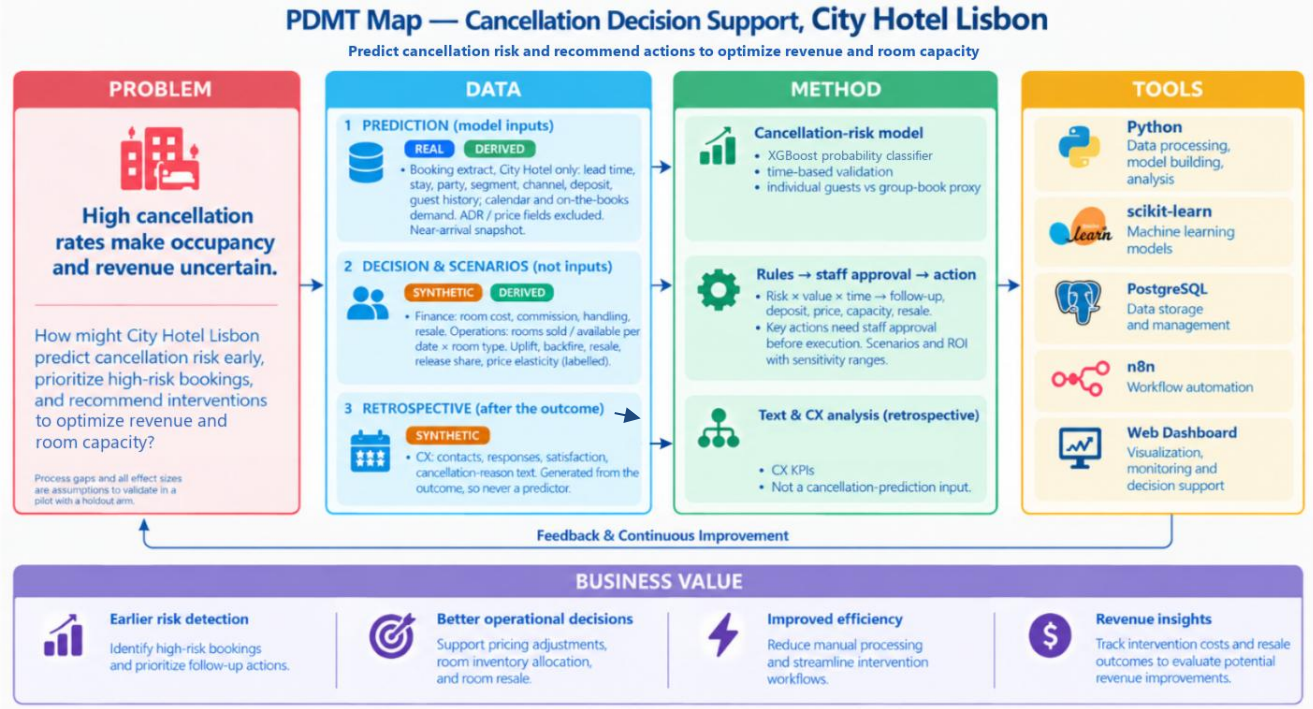
Targets are reference pilot goals, to be confirmed through model evaluation and a randomised pilot.

1.6. SCQ Framework

Situation	City Hotel Lisbon sells rooms in advance. Over July 2015 - August 2017, 41.7% of 79,330 bookings were cancelled or no-show, tying up €10.9M of €25.3M gross booked room revenue.
Complication	Most cancellations arrive weeks ahead (median 58 days), and 12.0% within 7 days, when resale is hardest. Risk differs sharply by segment (Direct 17.3%, Online TA 37.4%, Groups 68.9%), yet at-risk guests are not systematically contacted and cost is not linked to risk. With roughly 80 new individual bookings a day, manual review of every booking is not scalable.
Question	How might City Hotel Lisbon predict booking cancellation risk early, prioritize high-risk bookings, and recommend proportionate interventions to optimize revenue and room capacity?

2. PDMT Map

The map links the strategic problem (Section 1) to the data (Section 4), the two methods (Section 5) and the tools (Section 6), and to the business value expected from them (Section 3).



3. Expected Business Value

Baseline: City Hotel Lisbon, Jul 2015-Aug 2017 (41.7% cancellation rate). Targets without a sound benchmark will be set from model evaluation and baseline analysis. External benchmarks are pilot references from other settings.

Business function	Expected value and target	Evaluation and data
1. Cancellation risk prediction	Rank risk better than simple rules at the team's real capacity: Recall@20% at least equal to a segment × lead-time rule, Precision@20% ≥ 50% and Lift@20% ≥ 1.5 on a later, time-ordered test set (a 20% queue caps recall at about 58%).	Recall, Precision, PR-AUC vs. a baseline model on an independent test set. Data: is_canceled and booking-time features (lead_time, previous_cancellations, market_segment, distribution_channel, customer_type, agent); deposit_type and booking_changes are excluded to avoid leakage.
2. Risk-based prioritisation	Focus staff on the bookings with the most value at risk: each week, call the top 15% by risk × booking value and send reminders to the next 45%, within 20 call tasks a day.	Cancellation rate and share of cancelled value in the call and contact tiers, against the cancellation rate of the same weekly queue (2017: call tier 38.3% vs. 24.0%). Data: model risk score, booking value, is_canceled.

3. Customer experience and retention	Pilot target: a lower cancellation rate in the contacted tiers than in a randomised holdout, with no rise in opt-outs or complaints. Break-even is low: a call pays for itself if it prevents 1.1% of the cancellations it reaches, a reminder 2.0%.	Retention uplift (pp) = Retention(contact) − Retention(control); also response and cancellation rates. A control group is required for a causal claim. Data: contact_attempted, contact_channel, customer_response, is_canceled.
4. Finance and revenue recovery	Simulated net benefit of EUR 30.4k a year after variable costs and backfire losses; 3-year programme ROI of 173% and payback of 6.9 months from full roll-out (month 23 of the roadmap) after a EUR 15.6k one-off cost and EUR 3.4k a year running cost (Section 5.2). Assumption-based until the pilot measures the effect. Reference: about 4% hotel profit gain from revenue-management optimisation (4.20% in a simulated overbooking scenario; Dong & Ling, 2015; see also Pimentel et al., 2019). A model benchmark, not a guaranteed gain.	Net recovery = Rebooking revenue − Incremental costs (commission, handling, staff). The €10.9M gross figure is not net recovery. Staff-cost floor: Portugal minimum wage €505–557 per month, 2015–2017 (Hans-Böckler-Stiftung, n.d.).
5. Room inventory and operations	Reference: at least 2.3% higher contribution profit when decisions are updated over time rather than optimised once (2.27% weekly booking-limit update; Cha, 2026). A dynamic-optimisation benchmark, not a resale-rate benchmark.	Resale rate = Rooms resold after cancellation ÷ Cancelled rooms eligible for resale; also unsold room-nights by arrival date. Compare operations with and without cancellation prioritisation.

4. Data

The data combine one real PMS extract with simulated tables that fill its gaps. Every field is labelled REAL, DERIVED (computed from real data), SYNTHETIC (simulated from a documented assumptions register) or SYSTEM (written later by the model, rules and n8n), so simulated figures are never presented as facts.

4.1 Data sources and audit

- **Source:** City Hotel Lisbon bookings from a hotel PMS extract (Antonio et al., 2019): 79,330 of the file's 119,390 rows, arrivals July 2015 - August 2017, values recorded the day before arrival. The target is_canceled covers cancellations (40.6%) and no-shows (1.2%). The data are anonymised and match the published description on every check.
- **Audit:**

Issue	Finding	Decision
Missing values	company 95.4% and agent 10.3% empty (= none); country 24 rows, children 4 rows	Flag or fill
Duplicates	25,902 rows (32.7%), mostly Groups; cancel 66% vs 30%	Likely block bookings: flag, keep
Invalid values	One ADR of EUR 5,400; 700 paid bookings with ADR 0; 331 zero-night and 167 zero-guest rows	Flag; impossible rows excluded

- **Preliminary EDA:** the cancellation rate rises from 12% for bookings made 0–7 days ahead to 71% beyond a year; median notice is 58 days, so many cancelled rooms can in principle be resold.
- **Data gaps:** The data predict well but lack what revenue decisions need: 12 of 17 data needs are missing, mainly room cost, commission, inventory, resale and guest contact. These are filled with labelled synthetic tables (Section 4.3); intervention uplift can only come from a pilot.

4.2 Data architecture and schema

All tables share one key, booking_id: booking-level tables join 1:1, and room inventory joins through the nights each booking covers.

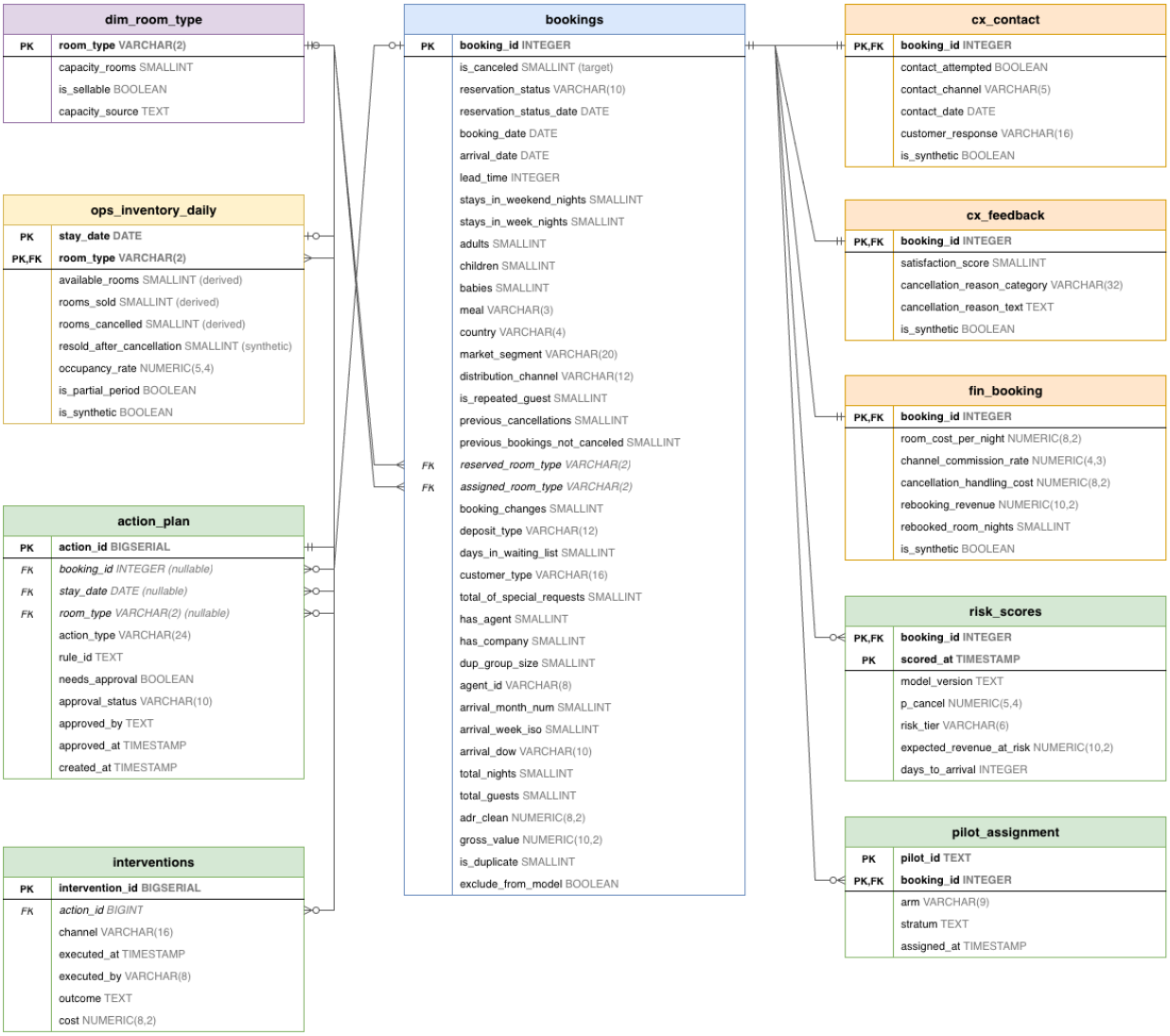


Figure 4.1. Data architecture

Simulated tables feed value-at-risk and scenarios, never the classifier. The schema runs in PostgreSQL with keys and CHECK constraints (for example, an action needing approval cannot run automatically) and passes 30 of 30 load checks; all 100 fields are documented in a data dictionary.

4.3 Data sources and features

- **Simulated data:** All values come from one assumptions register, sourced for Lisbon or Portugal 2015-2017 where possible, each with a sensitivity range.

Field	Base value	Basis

Room cost per night	27% of median ADR (room type × month)	Mandelbaum & McDade, 2017; Rushmore, 2011 (proxy); range 25-30%
Channel commission	Online TA 18%, Offline TA/TO 20%, Groups 10%, Direct 0%	Online TA: Toulon (2014), weighted by OTA shares (Schegg, 2018). Offline TA/TO and Groups: team assumption.
Handling cost	8-15 minutes × EUR 9.40 per hour	("Salários médios sobem," 2018)
Deposit retained	0% (Non Refund tested up to 100%)	Team assumption: amount not recorded
Resale	10-70% by notice × demand, at 90% of ADR	Team assumption; capped by rooms actually sold
Room capacity	252 rooms (peak occupancy per type)	Derived; 71.3% occupancy in 2016.

On these assumptions, 50.7% of cancelled room-nights are resold, and cancelled bookings lose about EUR 3.8M in contribution net of resale. The loss falls to EUR 0.6M if Non Refund deposits are kept, so the deposit assumption matters most.

- **Model features:** The individual model uses 39 booking-time features (32 numeric, 7 categorical): lead time, season, stay, party, guest history, top-15 agent and bookings on the books. Outcome fields, assigned room, country, booking_changes and price fields (ADR and its derivatives) are never used as features. Feature statistics are fitted on each training fold only.

4.4 Data preparation

Cleaning keeps all 79,330 rows and flags problems instead of deleting them. Only 438 impossible rows (0.55%: zero guests or nights, the ADR error, undefined segment) are excluded, leaving 78,892 for modelling. Duplicates are kept and flagged, and results are reported with and without them. The simulated tables pass 39 validation checks and are joined 1:1 with the prefix syn_.

Bookings are split into two populations because they need different decisions:

Population	Rule	Bookings	Cancel rate: 2015-16 → 2017
Individual	All other bookings	52,676	28.0% → 34.4%

Group / block	Groups segment, Non Refund deposit, or 5+ identical bookings	26,216	64.2% → 66.7%

Data are split by arrival date: 2015–2016 for model development (rolling-origin folds within 2016) and January–August 2017 for testing.

4.5 Exploratory analysis and leakage checks

- **Strongest signals:** deposit type, agent, segment and lead time. Among individual guests the signal is weaker (special requests, last-minute booking, lead time); season matters less (monthly rates 37–46%).
- **Group blocks** cancel early (median 83 days before arrival) and in batches; individuals cancel later (36 days) and one by one.
- **Drift 2016 → 2017:** new agents (PSI 1.09) and median ADR +16.5%, so agents are grouped as top 15 + others (rebuilt in each training fold) and price fields are left out of the model.
- **Leakage:** Fields not known at scoring time are excluded: reservation status, booking_changes, assigned room type, parking requests, realised occupancy and all simulated fields. 'Country' is excluded for fairness and because its booking-time value is unverified. EDA saw 2017 labels, so model choice and thresholds use the 2016 folds only.

4.6 Example data extract

A 30-booking stratified extract marks the origin of every column (real__, drv__, syn__). Three rows:

Case	Segment, deposit	Gross value (EUR)	Room cost per night (syn, EUR)	Commission (syn)	Resale revenue (syn, EUR)
Stayed, contacted by email	Online TA, No Deposit	324.90	25.34	18%	-
Cancelled, resold	Online TA, No Deposit	535.50	25.49	18%	200.37
Cancelled, not resold	Groups, Non Refund	192.00	24.70	10%	0.00

4.7 Integration challenges

Challenge	Design response

Several systems (PMS, channel manager, email/SMS, accounting) with different IDs	One PostgreSQL schema keyed on booking_id, plus an ID mapping table
Booking-level vs night × room-type data	Separate tables joined through stay nights
Snapshot data vs live scoring	Time-stamped scores and actions; production keeps booking-change history
Real-time inventory	Production: refreshed by n8n from the PMS. Prototype: synthetic table loaded in batch.

4.8 Privacy and ethics

The public data are anonymised; a live system would fall under the GDPR, so the design adds four safeguards:

- **Minimisation:** the model uses no names or contacts; personal data stay in a separate, access-controlled table.
- **Human review:** change offers, deposit requests, and room or price changes need staff approval (General Data Protection Regulation, 2016, Art. 22).
- **Fairness:** country is excluded from the model, and actions are monitored by segment and channel so no guest group is unfairly burdened.
- **Transparency:** guests are told that bookings are risk-scored, and reports label every simulated figure.

5. Method

5.1. Method 1: Cancellation Risk Prediction (ML)

5.1.1. Purpose and scope

Purpose: Score each individual booking by its cancellation risk at an early stage, so staff can prioritise a limited number of bookings for follow-up. The model ranks bookings. It does not make the final decision. The ranked list feeds Method 2 (rules → staff approval → action).

Scope:

- Individual bookings only. Group bookings (group_block) follow a separate operational rule.
- Target: is_canceled (cancelled or no-show).
- Data: real PMS extract, City Hotel Lisbon, Jul 2015 - Aug 2017.

5.1.2. Analytical design

Flowchart:

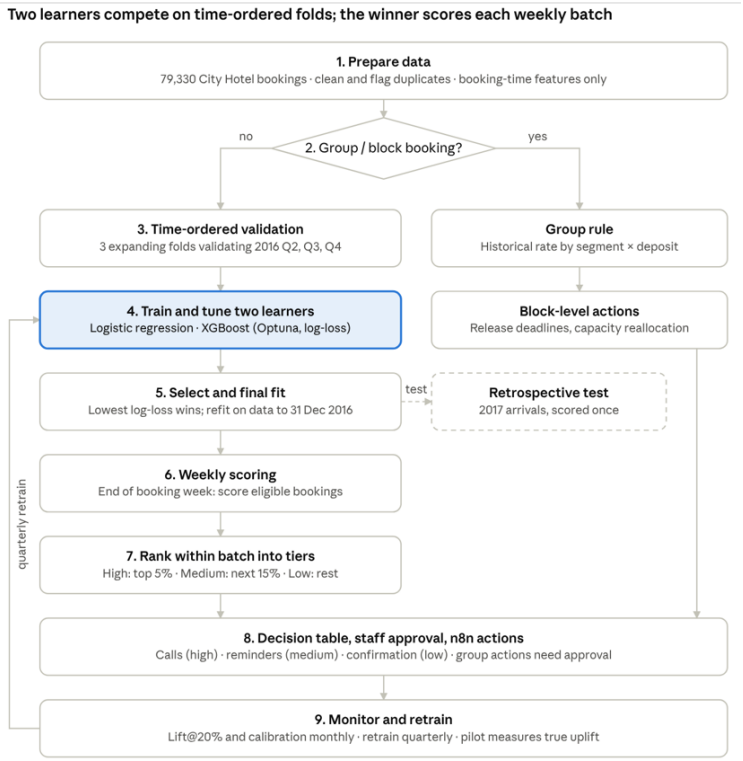


Figure 1. Method 1 analytical pipeline

Model selection rationale:

We select models by how often their families appear in hotel-cancellation research and by what each does in a binary classification problem (Alkan, 2025; Chen et al., 2022; Hermawan et al., 2025; Luo, 2025; Yoo et al., 2024). XGBoost and Logistic regression appear in most of the studies. What each model means for this classification problem:

- XGBoost: captures non-linear effects and interactions such as agent × lead time, and handles mixed tabular data.
- Logistic regression (benchmark): gives odds ratios managers can read and stable probabilities

These two models represent a linear and a boosting approach. They suit our data: 79,330 bookings, mixed numeric and categorical features, threshold effects (lead time) and interactions (deposit × segment). Both output probabilities, which the weekly ranking into tiers requires. The dataset is small enough for every model to receive the same tuning budget, so the comparison is fair.

5.1.3. Prototype analysis (mock results)

On 2017 data the model ranks risk clearly better than the hotel's best simple rule; results are retrospective and the weekly queue is simulated.

Scorer	AUC	PR-AUC	Brier	Precision@5%	Precision @20%	Lift @20%
XGBoost	**0.736**	**0.556**	**0.192**	**66.2%**	**58.3%**	**1.70**
Rule: segment × lead	0.678	0.468	0.206	54.1%	50.8%	1.48

time						
Logistic regression (lead time + segment)	0.676	0.493	0.209	61.5%	50.9%	1.48
Random / constant	0.500	0.344	0.230	34.4%	34.4%	1.00

- Gain over the rule: +308 cancellations caught in the same top 20% (bootstrap CI for precision: +5.4 to +10.0 points).
- Ranking: the top decile cancels 60.7%, 8.5 times the lowest (7.2%); ranking by expected loss puts 52% of cancelled revenue in the top 20% (random: 20%).
- Drivers: travel agent, lead time, customer type, length of stay (associations, not causes).

Simulated weekly queue 2017 (12,102 eligible bookings): High tier (4.9%, about 4 calls/day) cancelled 46.2%, Medium (15.0%, reminders) 39.4%, Low (80.1%, standard confirmation) 19.8%. At equal capacity the model catches 139 more cancellations than the rule (95% CI 48–232). These are Method 1's default tiers (5% / 20%); the operating policy uses the G1 tiers of Method 2 (top 15% called, up to 60% contacted, Section 5.2).

5.1.4. Evaluation plan

The method is evaluated in three steps: model accuracy (done offline), queue performance (simulated, then shadow-run), and business impact (randomised pilot).

KPI	Target	2017 result
Recall@20%	≥ rule	33.9% vs 29.5%
Precision@20% / Lift@20%	≥ 50% / ≥ 1.5	58.3% / 1.70
PR-AUC, Brier	better than baseline LR	0.556, 0.192 vs 0.493, 0.209
Revenue captured@20%	≥ 50% (expected loss)	52.2%

- Validation: time-ordered folds only, leakage controls inside each fold, one test pass, paired bootstrap against rules; robust to duplicates and group-rule changes (AUC 0.735–0.740).
- Monitoring: monthly Lift@20% and predicted vs actual rate. The model is frozen during the pilot so that treatment and holdout are compared under one policy; quarterly retraining starts at roll-out, since AUC fell from 0.80 to 0.71 during 2017. The dashboard's Model monitoring page recomputes these figures on the bookings that arrived before today (on 1 May 2017: 4,315 bookings, ROC-AUC 0.740, Lift@20% 1.92), not on notebook results that include later arrivals.
- Pilot: randomised treatment vs holdout; compare stay rate and net contribution by assigned arm, with complaint guardrails; scale up only if uplift is positive.

Limitations:

- Near-arrival snapshot data. Lead time, number of guests and room type are final values, not the values at the time of booking. All results are therefore retrospective, including the simulated weekly queue. They do not show how a live system would perform. Operational testing requires a snapshot or event log captured at scoring time.
- The group_block definition is inferred. It is not a label available in the data.

- Limits of the simulated queue. The queue includes only bookings created in weeks ending after 31 Dec 2016. There is no data on consent, opt-out or open actions. Summer weeks are missing bookings for arrivals after 31 Aug 2017.
- Summed probabilities by arrival week are not an operational room forecast.
- Detection is not prevention. How many cancellations and how much revenue are actually "retained" depends on how well the interventions work. That figure must be measured through a randomised pilot with a holdout group (pilot_assignment). It cannot be inferred from simulated or synthetic contact data.

5.2. Method 2: Risk-Based Decision Rules and Business Value

5.2.1. Purpose and scope

Purpose: Convert the Method 1 risk score into weekly staff actions and test whether they pay off: which bookings to act on, what to do with each, and the net gain after all costs.

Scope:

- **Units:** individual bookings scored by the final Method 1 model (XGBoost). Group blocks follow separate rules (G1–G3), all approved by the Revenue Manager.
- **Decision time:** end of each booking week, covering bookings created that week that are still active and not yet arrived.
- **Evaluation:** 2017 operational queue (12,102 individual bookings, 35 weekly batches, 24.0% cancelled). The policy was chosen on 2016 data and frozen before 2017 was read.
- **Money:** contribution basis (room revenue net of commission, minus room cost). Uplift, cost and resale inputs are assumed or simulated (Section 4), not observed.

5.2.2. Analytical design

Step 1: Tier each weekly batch. The **call tier** is the top 15% by p_cancel × booking value, so calls target the most money at stake. The **reminder tier** takes the next bookings by p_cancel up to 60% coverage. The **low tier** is the remaining 40%. Shares and ranking key were chosen from 42 candidate policies on 2016 batches, using models fitted only on data available before each decision. Batch shares, unlike a fixed cut-off, keep workload stable when scores drift.

Step 2: Route each booking to an action (Figure 5.2):

Rule	Condition	Action	Owner	Approval
I1	Low tier, or reminder tier ≤ 7 days to arrival	Standard confirmation + reminder 7 days before arrival (current practice)	n8n, monitored by Reservations	Weekly plan review, then automatic
I2	Reminder tier, ≥ 8 days	Personalised e-mail/SMS at 30 and 14 days (max 2)	n8n, monitored by Reservations	Weekly plan review, then automatic

I3	Call tier, ≥ 8 days	Call within 7 days of decision, ≥ 8 days before arrival; max 20 tasks/day; unslotted tasks → manual review	Reservations	Weekly plan review (internal task)
I4	Guest requests change during I3 call	Offer date or room change	Reservations	Revenue Manager (every offer in the prototype)
I5	Call tier, ≤ 7 days	No contact; count in date × room-type cancellation buffer	Revenue Manager	–
I6	Call tier, no deposit, high value	Request deposit (off by default)	Reservations	Revenue Manager (GDPR Art. 22)
G1	Group block, risk ≥ 0.5, > 30 days ahead	Ask agent to confirm or release block 30 days before arrival (valued in scenario S5; not yet an action in the prototype)	Distribution / Sales	Revenue Manager
G2	Released or forecast-cancelled rooms	Reallocate capacity; return rooms to sale	Revenue Manager	Required
G5	Price change on a date × room type	Adjust price (no tested price limits yet)	Revenue Manager	Required
G3	High forecast cancellations on high-demand date	Controlled overbooking (disabled; exploratory only)	Revenue Manager	GM sets limits

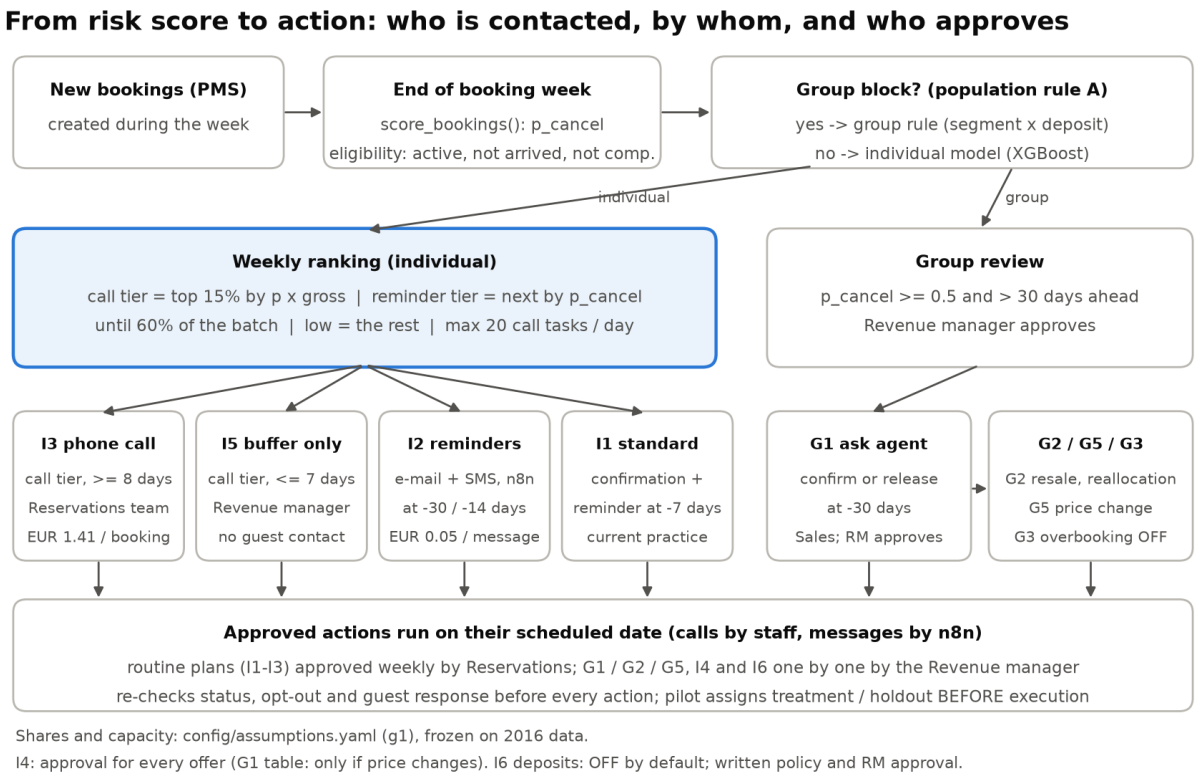


Figure 5.2. Decision workflow: from risk score to action

In production, n8n runs automatic actions on schedule after re-checking booking status, opt-out and guest response. In the prototype (Section 6), these checks, the call window and the daily limit run in the dashboard and in a demo n8n workflow that sends nothing. Actions affecting price, capacity or guest obligations require approval.

Step 3: Value each action.

- **Benefit:** for a booking that would have been cancelled, contact prevents a share of cancellation (uplift). Each prevented cancellation is worth its contribution minus what would have been recovered anyway (resale contribution − handling cost + any deposit kept). Late cancellations are worth more, as rooms are harder to resell.
- **Backfire:** since 99.96% of individual bookings are No Deposit and free to cancel, a reminder may trigger a cancellation that would not otherwise occur. This loss is subtracted.
- **Costs:** EUR 1.41 per call task (6 min × 1.5 attempts at EUR 9.40/h) and EUR 0.05 per SMS, with none for bookings cancelling before contact. Programme: EUR 15,564 one-off (integration, n8n workflows, dashboard, training) + EUR 3,360/year (hosting, monitoring).
- **Base assumptions:** uplift 10% (calls) / 3% (reminders); backfire 0% / 0.5%. Reminder uplift is set well below the SMS effect on healthcare no-shows (Gurol-Urganci et al., 2013), as hotel cancellations are mostly deliberate.

5.2.3. Prototype analysis (mock results)

Scenario comparison (2017 operational queue, simulated, annualised):

Scenario	Calls	Reminder bookings	Cancellations prevented	Net benefit per year*
S0 Status quo	0	0	0	0
S1 Reminders for all	0	10,703	39	EUR 3.1k
S2 Rule-based tiers (5% / 20%)	586	1,810	27	EUR 4.4k
S3 Model tiers, Method 1 default (5% / 20%)	574	1,803	35	EUR 12.6k
S4 G1 policy (15% / 60%, proposed)	1,744	5,186	86	**EUR 30.4k**
S5 S4 + group track (G1 release asks)	1,744	5,186	86	EUR 31.5k

After variable intervention costs and backfire losses; before programme costs. Prevented cancellations are expected values over the 35-week queue.

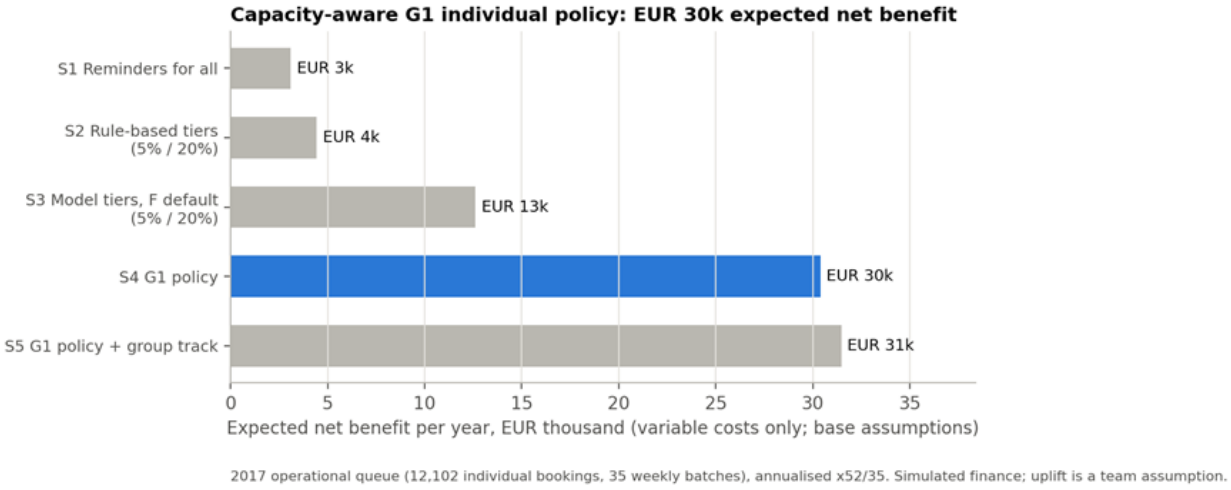


Figure 5.3. Net benefit per year by scenario (2017 queue, simulated)

Programme view (S4, 3 years of full operation from roll-out, undiscounted): year-1 net EUR 11.5k; 3-year net EUR 65.5k; ROI 173%; payback 6.9 months after roll-out. During the pilot only the treatment arm (half of eligible bookings) is contacted, so benefits before roll-out are smaller and are measured, not assumed.

Key findings:

- **Value targeting beats contacting everyone.** S1 earns only EUR 3.1k because backfire on low-risk guests nearly offsets the benefit. S4 earns ten times more.
- **The model beats a simple rule, modestly.** At the same shares it earns EUR 30.4k vs. EUR 28.8k for a segment × lead-time rule.
- **Call uplift drives the case.** Break-even uplift is 1.1% for calls and 2.0% for reminders. At zero call effect, S4 loses EUR 1.9k/year (programme ROI –82%). At 25%, net reaches EUR 78.8k.

- **Other assumptions matter less.**
  - Reminder uplift 0% / 8%: EUR 25.5k / 38.5k.
  - Resale 0.5× / 1.5×: EUR 40.3k / 20.5k.
  - Reminder backfire 2%: EUR 22.4k.
  - Call backfire 1%: EUR 26.5k.
  - Implementation cost EUR 30k: ROI 95%, payback 13.6 months.
  - Call time, attempts and SMS price: each < EUR 3.6k.
- **Capacity holds.** Calls average 16.5 per scheduled day (max 20). 17 tasks go to manual review.
- **Overbooking (G3) stays off.** It loses EUR 0.7k/year after relocation costs and is not recommended without a new capacity design.

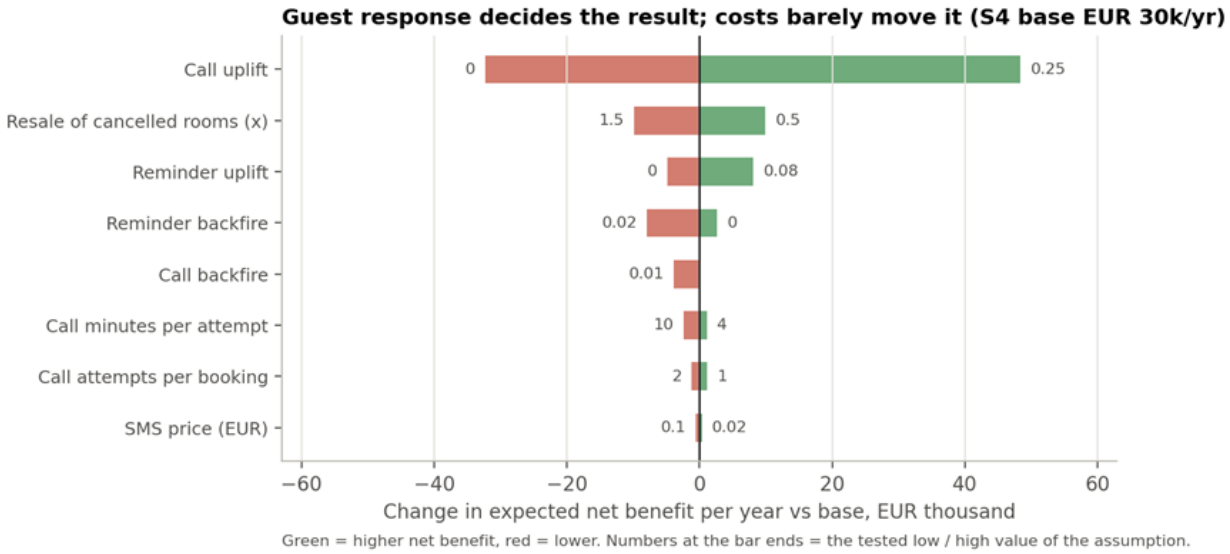


Figure 5.4. Sensitivity of S4 net benefit to each assumption

Occupancy, ADR and RevPAR (room-night ledger, stays 1 January – 31 August 2017, illustrative):

Scenario	Occupancy	ADR (EUR)	RevPAR (EUR)	Contribution net*
S0 Actual 2017	78.7%	116.96	92.02	–
S4 G1 policy	79.0%	117.22	92.59	EUR +20.6k
S5 S4 + group early release	79.0%	117.20	92.63	EUR +21.3k
S6 S5 + price ±5% (elasticity −0.8, assumed)	76.8%	121.37	93.16	EUR +85.6k

After variable costs, commission and room cost; before programme costs. Expected values from src/scenario_capacity.py, based on the 48,182 of 61,236 room-nights sold. The S4 result matches the queue simulation (EUR 20.5k over 35 weeks). S6 raises rates 5% on dates ≥ 85% occupancy

and cuts them below 60%. Its elasticity is a team assumption (no price variation in the data), and at −1.2 or −1.5 RevPAR falls, so S6 is a pilot question, not a recommendation. Overbooking (G3) is not simulated as an option.

5.2.4. Evaluation plan

Level	Measure	How	Success criteria	Current status
Queue control	Queue size, call tasks/day, manual reviews	Action schedule, intervention log	≤ 20 calls/day; manual review rare	Max 20/day; 17 tasks reviewed (2017)
Targeting quality	Cancellation rate and value share in contacted tiers	Method 1 weekly queue	Lift > 1; model ≥ rule	Call tier: 38.3% vs. 24.0% cancelled (lift 1.60), 44.8% of cancelled value. Call + reminder: 87.4%. Model top 20% precision 41.1% (rule 35.3%)
Business value	Net benefit, ROI, payback, break-even uplift	Scenario model, one-way sensitivity	Positive net under assumptions	S4 EUR 30.4k/year; ROI 173%; payback 6.9 months
Causal effect	Stay rate and contribution, treatment vs. holdout	Randomised pilot, assigned before contact, stratified by tier; analysed by assigned arm	Lower cancellation in treatment, with CI	Not yet available
Guest experience	Opt-outs, complaints, post-reminder cancellations (backfire)	CX monitoring by arm	No deterioration vs. holdout	Not yet available

Pilot sizing: a one-year, one-hotel pilot, followed by about four months until enrolled bookings reach their arrival date, can detect a reduction of about 5 percentage points as a stop / redesign / scale decision with CX guard-rails, or be pooled across more properties, not as proof of a precise effect.

Limitations:

- Uplift, backfire, cost and resale values are assumed or simulated, so results show logic and sensitivity, not measured return.
- The policy was chosen using a model selected on 2015 data and refitted before each 2016 decision, then applied to the final Method 1 model. The pilot must confirm the transfer.
- About 4,600 bookings in the 2016 selection queue had no outcome at the freeze date, under-representing very long-lead bookings.
- Snapshot features and the group-block proxy come from Method 1, so 2017 results are a retrospective prototype.

Output to the system (Tools): a weekly action queue and dated schedule per booking (booking ID, risk score, tier, rule, action, planned date, approval need), with the 17 unslotted call tasks flagged for manual review. Staff approve each week's routine plan before any contact is scheduled; n8n runs automatic actions in production, while in the prototype a demo workflow only checks and logs them. Approval-dependent actions wait for the Revenue Manager.

6. Tools and System Prototype

The prototype links booking-level risk estimation (Method 1), the capacity-aware decision policy (Stage G1) and an operational layer for staff: a web dashboard, an approval-guarded action store and an n8n workflow. It works on one fixed day, today = Monday 1 May 2017 (set at start with G2_TODAY), as staff would meet it in a pilot: the weeks before it have been handled through the same service and guards, and the e-mails and reminders of the batch scored yesterday wait for today's review (its calls were approved yesterday, as calls start on the scoring day). The dashboard, action store and n8n workflow are implemented and tested, but no guest is contacted and no PMS, price or inventory record is changed. Section 6.5 separates what is built from what remains a production design.

6.1. Technology Stack

The stack separates prediction, decision rules, data management and delivery:

- **Python** produces scores, tiers and dated actions.
- **CSV/JSON files** pass them, with a run identifier, to the operational layer.
- **PostgreSQL** holds the relational data model and, optionally, the shared action store.
- **Streamlit** provides the staff interface.
- **n8n** shows how an approved action is handed to a workflow.

Table 6.1 assesses each tool against the brief's three criteria.

Tool	Role	Scalability	Collaboration	Usability for managers
Python (pandas, scikit-learn, XGBoost)	Weekly scoring (F), G1 tiers, capacity-limited schedule, scenario analysis	Weekly batch of new bookings;	Run ID and code/input hashes in each manifest	Invisible; results reach managers via the dashboard

		reproducible runs		
CSV / JSON contract	Hands risk scores, action queue, dated schedule and G1 policy to the dashboard	Plain files now; same fields can move to tables or an API	Human-readable, versioned with code	Opens in a spreadsheet for review
PostgreSQL 16	Hotel data model (10 tables, 2 views); shared action store with the same rules as SQLite	Constraints, indexes, multi-user access; writes serialised per store	One shared source for Reservations, Revenue Management and Finance	Managers never write SQL
SQLite	Local action store (default if no PostgreSQL address is set)	Single machine only	Not shared; one file per model run, policy and day	No set-up for a first demo
Streamlit	Dashboard: overview, risk queue with batch review, action review and approvals, execution and follow-up, capacity planning, performance, model monitoring, data scope	Fine for a pilot; needs login and server deployment at scale	Python code in the same repository	Filters, tables and buttons; no coding
n8n	Demo workflow: webhook, approval and execution guard, mock dispatch, reply	Workers and queues can be added in production	Visual nodes reviewable by operations staff	Readable sequence of steps
unittest + Node.js	Tests of data contract, guard rules, concurrency, workflow code and app	Full suite runs in about four minutes	Same checks for every team member	Not used by managers

Table 6.1. Technology stack, role in the prototype and assessment

6.2. Data Integration and Analytical Outputs

The selected XGBoost model (f-v3.1-2026-10-19) scores bookings created in each booking week that are still active at its end. For 2017 this gave 13,156 risk-score records: 12,102 individual bookings (XGBoost) and 1,054 group-block bookings scored by the group rule (segment × deposit), which go to the capacity track. Each record holds the booking ID, scoring time, model version, cancellation probability, risk tier, expected revenue at risk and days to arrival. G1 then re-tiers the individual bookings into a dated, capacity-limited action schedule (Table 6.2).

Output	Recorded result	Operational use
Risk scores	13,156 records (12,102 individual, XGBoost; 1,054 group rule)	Cancellation probability and revenue exposure per booking
Decision queue (G1)	12,102 bookings: I1 = 5,116; I2 = 5,186; I3 = 1,761; I5 = 39; 17 unslotted call tasks → manual review	Rule, tier and owner per booking
Action schedule	20,563 dated actions: 5,116 confirmation e-mails, 13,703 reminders, 1,744 call tasks	Daily due list, imported into the action store once staff approve the weekly plan
Decision policy	Calls: top 15% per weekly batch by risk × value; contact: 60% of batch; ≤ 20 calls/day, within 7 days of decision and ≥ 8 days before arrival; G3 overbooking disabled	Keeps contact within staff capacity
Database structure	Hotel model: 10 tables, 2 views, checked by SQL scripts (30/30 pass). Action store (schema 9): 8 tables with guard triggers, on SQLite or PostgreSQL	Bookings, scores, actions, approvals, interventions, guest responses

Table 6.2. Analytical outputs used by the prototype (2017 evaluation period)

Before showing anything, the dashboard runs four checks:

(1) It confirms that the model and policy files come from the same run.
(2) It re-derives every G1 tier from the scores.
(3) It validates each dated action against the policy.
(4) It rejects any field known only after the outcome.

Outcomes (cancelled or not) are read only for bookings that arrived before today, to show what happened (Overview, Performance, Model monitoring) and to keep the prepared guest answers consistent; they never feed the queue, the schedule or a recommendation. Monetary figures shown

to staff (expected revenue at risk; net value at risk on simulated costs) are scenario estimates, not realised or recovered revenue.

6.3. System Architecture and Workflow

The architecture follows the flow from booking data to risk estimation, decision policy and staff action (Figure 6.1). Analytical stages run as weekly Python batches and hand results over via the file contract. Staff work in the dashboard. Every proposal, approval, execution, guest response and closure is written to the action store, whose rules are enforced in both the service and database triggers. n8n receives an action only after these checks pass.

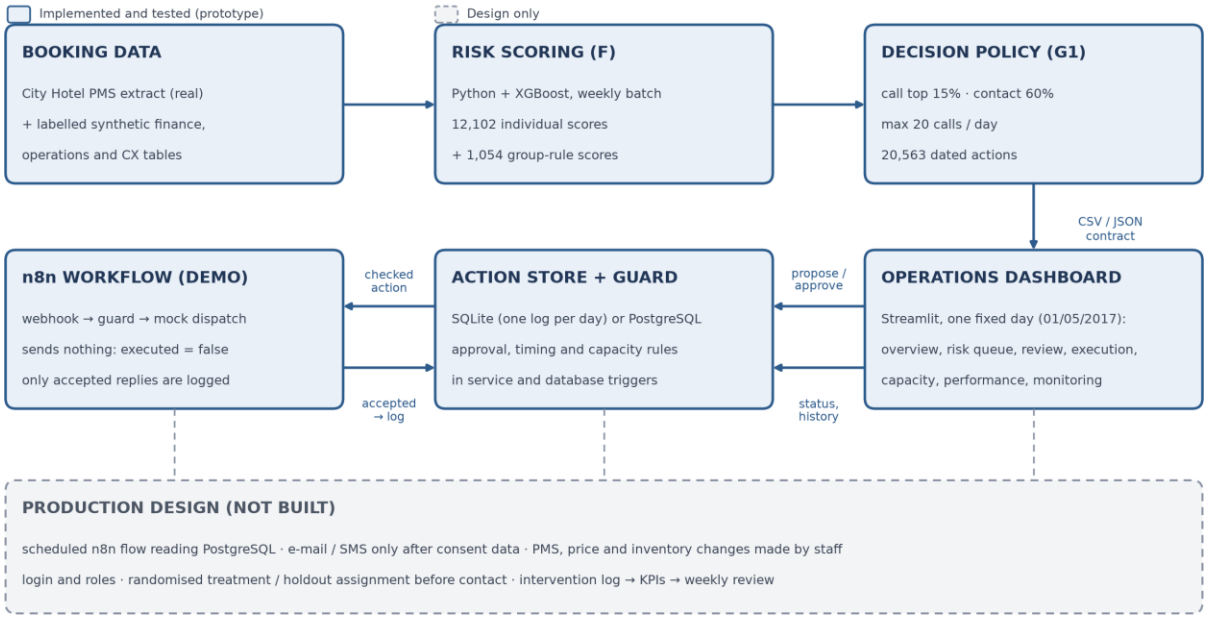


Figure 6.1. Technical architecture and booking-level decision workflow (solid: implemented; dashed: production design).

Before execution, the store checks that the action:

- is approved (or automatic);
- is not already executed or closed;
- is due today and falls before arrival.

It also checks that a call is still within the G1 window and the 20-per-day limit, that the guest has not already responded, and that no automated message goes out on the same day as a call. Consent, other open actions and whether the booking is still active are not in the data, so staff must confirm them for each guest action.

The checks, the n8n call and the log entry form one database transaction. As a result, concurrent staff cannot exceed the call limit or both close and execute the same action. Logs are append-only.

6.4. Dashboard and Automation Process

The dashboard centres on the day's work, not model diagnostics. It has eight pages, grouped as daily work, planning and analysis, and data:

(1) Overview: yesterday or the last 7 days (bookings scored, high risk, approvals, contacts, confirmations, arrivals, cancellations and how many the model had flagged), then today's work (Figure 6.2).

(2) Risk Queue: open bookings filtered by risk level, review and execution status and dates; routine recommendations approved in a batch, with a summary and a confirmation step.
(3) Action Review: one booking in depth; a recommendation rejected with a reason or an action replaced; sensitive actions approved one by one (Figure 6.3).
(4) Execution & follow-up: what is due today, calls and guest answers recorded, automation that refused or failed (with Retry), actions closed without contact, each booking's history.
(5) Capacity planning: expected cancellations per stay date × room type; room or price actions proposed for approval.
(6) Performance: operations so far and cancellations of the bookings already arrived (descriptive).
(7) Model monitoring: ROC-AUC, PR-AUC, Brier, top-20% precision, recall and lift, and deciles, recomputed on the bookings arrived before today.
(8) Data scope: sources, load checks, coverage and the fields that are not in the data.

Every page works from one fixed day, shown in a Today band with the latest scores, calls logged out of 20, actions planned for today and actions waiting for approval; nothing can run before its due date. Review status (has staff decided on the recommendation?) and execution status (has the action been carried out?) are kept apart, and nothing shows as sent or completed without a logged execution.

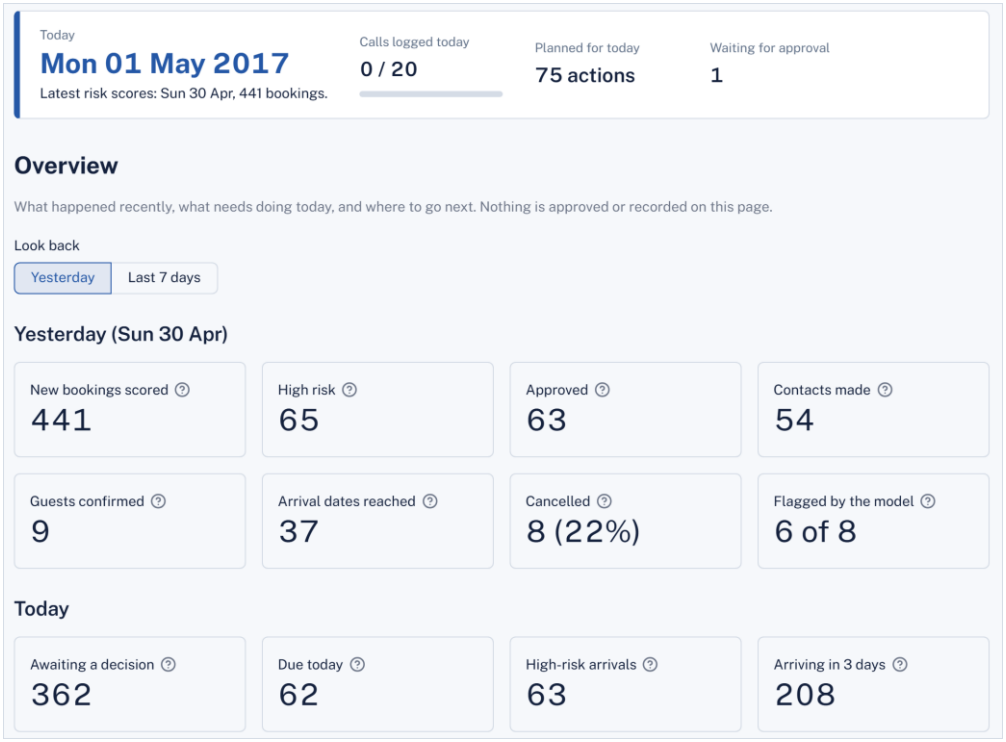


Figure 6.2. Overview on today, Monday 1 May 2017: yesterday's batch (441 bookings scored, 65 high risk, 63 calls approved) and today's work (362 decisions waiting, 62 actions due).

Prepared history. A new action log first receives the team's work up to yesterday, written once through the store's own service so every guard applies: routine recommendations approved on their scoring day, contacts carried out on their due date (e-mails and reminders through automation, calls by staff, at most 20 a day), follow-ups closed once a guest has answered, and one room-hold

action a week approved by the Revenue Manager. The data hold no guest answers, so they are derived from each booking's real outcome and never contradict it: a guest who later cancelled never confirms. With today = 1 May 2017 this gives 9,039 approved recommendations, 1,332 calls and 8,876 e-mails and reminders.

Rule	Operational handling	Control measure
I1 – Standard	Confirmation on decision day + reminder 7 days before arrival (current practice)	Weekly plan review, then automatic; stops once the guest responds
I2 – Reminder	Personalised e-mail/SMS at 30 and 14 days before arrival (one at once if a milestone has passed)	Max 2; never on a call day; stopped by any guest response
I3 – Call	Reservations call task within 7 days of decision, highest risk × value first	Max 20 calls/day, 2 attempts per booking; outside window: manual review or close without contact
I4 – Offer change	Date or room change after a call where the guest asked to change	Revenue Manager approval, tied to that request; a newer request needs a new offer
I5 – Buffer only	Call-tier booking ≤ 7 days to arrival: no contact, counted in the buffer	Revenue Manager reviews the buffer
I6 – Deposit	Deposit request for a high-tier, no-deposit booking	Off by default; needs written policy and Revenue Manager approval
G1 – Group release ask	Ask the agent to confirm or release a block 30 days ahead (valued in scenario S5)	Not yet an action in the prototype
G2 – Capacity and resale	Capacity allocation or resale per arrival date × room type	Pending until a named Revenue Manager approves
G5 – Price change	Price adjustment per arrival date × room type	Pending until a named Revenue Manager approves; no tested price limits yet

G3 – Overbooking	Not available	Disabled; refused by the service, n8n and database

Table 6.3. Intervention routing and control rules

Approval is enforced in the database as well as the interface. A sensitive action starts as "pending" and cannot run until a named reviewer approves it (Figure 6.3). Approval decisions cannot be reversed or edited.

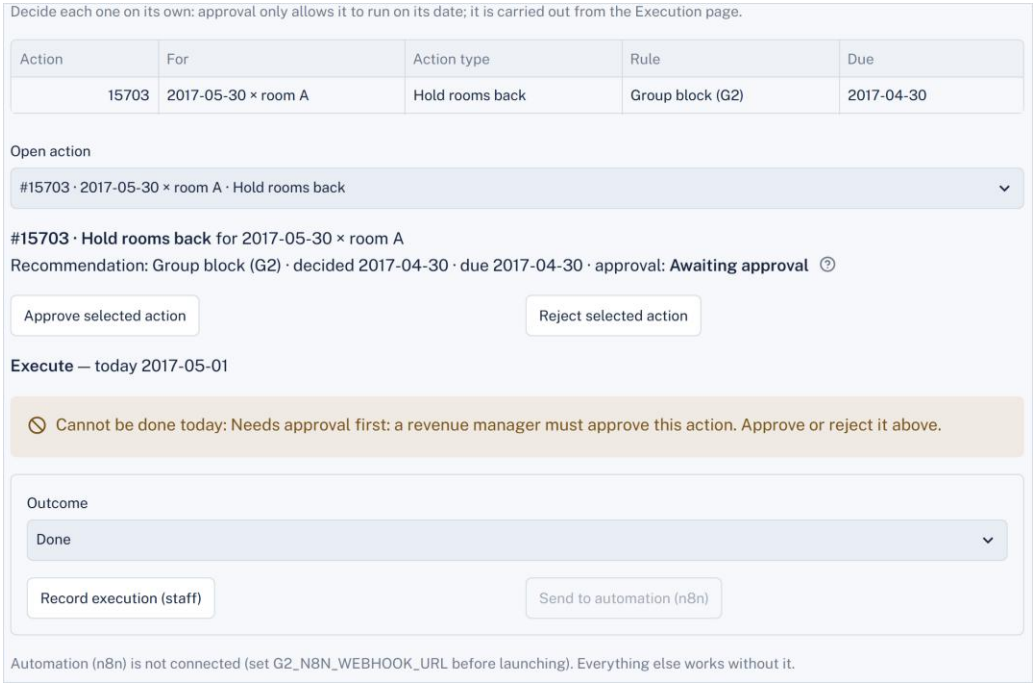


Figure 6.3. A pending capacity-allocation action (30 May 2017, room A) is blocked until the Revenue Manager approves it.

For automatic and approved actions, staff either log execution themselves or hand it to n8n. n8n re-checks approval and timing, prepares what it would send (an e-mail preview, staff task or approval notice) and replies. Only an "accepted: true" reply is logged, as executed by n8n (Figure 6.4). In this prototype nothing is sent ("executed": false).



Figure 6.4. A call task handed to the local n8n workflow (recorded on 20/10/2026, n8n 2.43.8): accepted, logged by n8n, nothing sent.

6.5. Verification and Limitations

Check	Result
Automated tests, SQLite store	263 tests pass (5 skipped: 2 need live n8n, 3 need PostgreSQL), 21/10/2026
Action-store tests, PostgreSQL 16	172 tests pass (4 skipped: 2 SQLite-file only, 2 need live n8n), 20/10/2026, before the history and monitoring tests were added
Live n8n workflow (local)	Approved call accepted and logged; pending action and overbooking refused by n8n's own guard (n8n 2.43.8, 20/10/2026)
Concurrent staff sessions	Last call slot goes to one session only; an action cannot be both closed and executed; each action reaches n8n once
Database scripts	Data load 30/30 checks; approval-guard migration 19/19 checks

Table 6.4. Verification of the prototype

Limitations:

- The prototype has no login or user roles, holds no guest contact details and contacts no one.
- Consent and open actions are not in the data, so staff must confirm them.
- The n8n workflow is a demonstration. In production it would run on a schedule, read due actions from PostgreSQL, send e-mail/SMS only once consent data exist, and leave PMS, price and inventory changes to staff.
- Pilot treatment/holdout assignment, needed to measure the real effect of contacts, is designed but not implemented.
- All financial figures remain scenario estimates.
- The history up to today is prepared, not observed: approvals, contacts and guest answers are generated by the system's own rules, and the answers are derived from each booking's real outcome.

7. Implementation Roadmap

We propose a 22-month path to a scale decision: five months to prepare and shadow-run the system, a 12-month randomised pilot, four months for pilot bookings to reach their arrival date, and one month to evaluate before the hotel commits to roll-out.

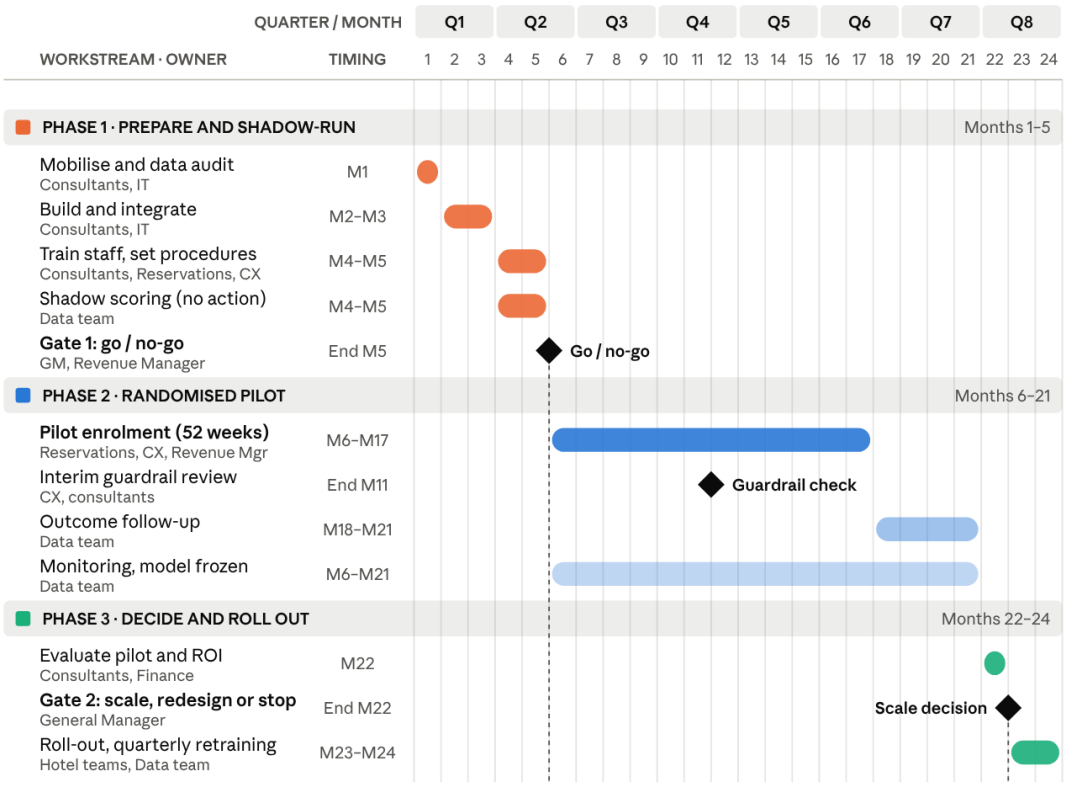


Figure 7. Implementation Roadmap for Lisbon

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